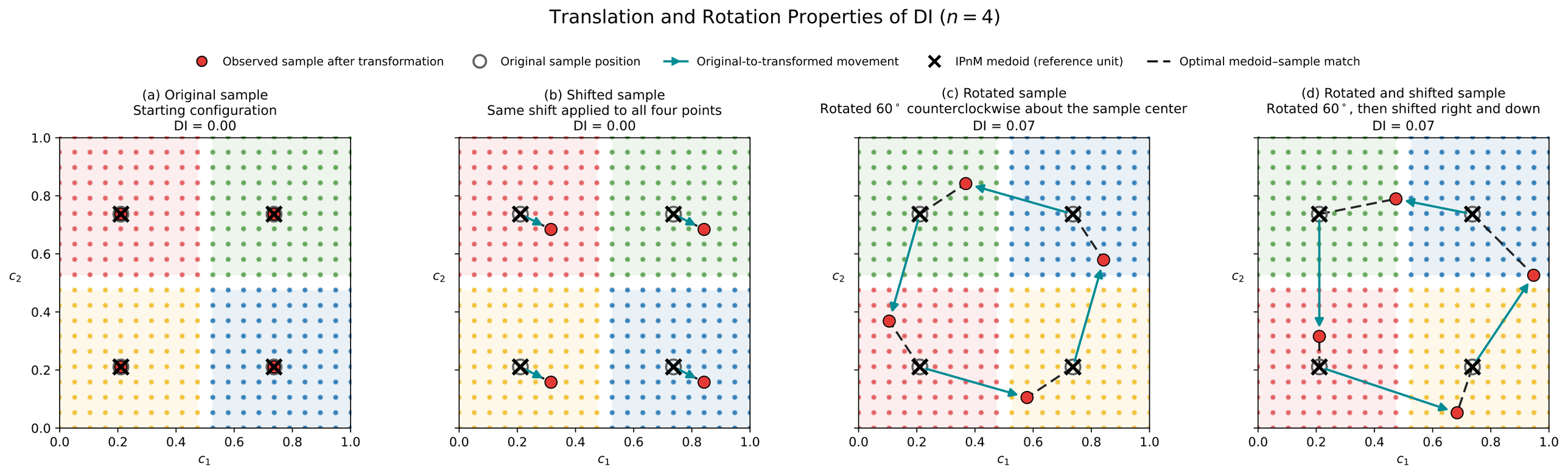

Translation and Rotation Properties of DI ($n = 4$)
Observed sample after transformation
Original sample position
Original-to-transformed movement
IPnM medoid (reference unit)
Optimal medoid–sample match
(a) Original sample
Starting configuration
DI = 0.00
(b) Shifted sample
Same shift applied to all four points
DI = 0.00
(c) Rotated sample
Rotated 60° counterclockwise about the sample center
DI = 0.07
(d) Rotated and shifted sample
Rotated 60°, then shifted right and down
DI = 0.07
c_1
c_2
0.0
0.2
0.4
0.6
0.8
1.0